

IGNOU BCS-012: Basic Mathematics
Term-End Examination, June 2024 - Fully Solved
BCA (Revised) Time: 3 Hours Maximum Marks: 100

Contents

Question 1	2
Question 2	5
Question 3	7
Question 4	9
Question 5	11
Last Page: Theory and Formula Sheet	13

Question 1

1(a) Show that

[5 marks]

$$\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$$

Answer. Let $s = a + b + c$ and

$$N = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The matrix on the left is

$$M = \begin{pmatrix} s-a & s-b & s-c \\ s-b & s-c & s-a \\ s-c & s-a & s-b \end{pmatrix} = sJ - N.$$

The vector $(1, 1, 1)^T$ is an eigenvector of N with eigenvalue s . Therefore for $M = sJ - N$, the eigenvalue of J on $(1, 1, 1)^T$ is 3, so the eigenvalue of M on this vector is $3s - s = 2s$. On the two-dimensional subspace $x + y + z = 0$, J acts as zero, so $M = -N$ there. Hence the three eigenvalues of M are $2s, -\lambda_2, -\lambda_3$, while those of N are s, λ_2, λ_3 . Thus

$$\det M = (2s)(-\lambda_2)(-\lambda_3) = 2s\lambda_2\lambda_3 = 2 \det N.$$

Final answer: The determinant identity is proved.

1(b) If $X = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ and $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find $(X - I_2)^2$

[5 marks]

Answer.

$$X - I_2 = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}.$$

Therefore

$$(X - I_2)^2 = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} = -4I_2.$$

Final answer: $(X - I_2)^2 = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix}.$

1(c) Find the sum up to n terms of $0.3 + 0.33 + 0.333 + \dots$

[5 marks]

Answer. The r th term is

$$\underbrace{0.\overbrace{33\cdots 3}^{r \text{ times}}} = 3 \left(10^{-1} + 10^{-2} + \dots + 10^{-r} \right) = \frac{1}{3}(1 - 10^{-r}).$$

Thus

$$S_n = \sum_{r=1}^n \frac{1}{3}(1 - 10^{-r}) = \frac{n}{3} - \frac{1}{3} \sum_{r=1}^n 10^{-r}.$$

Now

$$\sum_{r=1}^n 10^{-r} = \frac{1}{10} \frac{1 - 10^{-n}}{1 - \frac{1}{10}} = \frac{1}{9}(1 - 10^{-n}).$$

Hence

$$S_n = \frac{n}{3} - \frac{1}{27}(1 - 10^{-n}).$$

Final answer: $S_n = \frac{n}{3} - \frac{1}{27}(1 - 10^{-n}).$

1(d) If 7 times the 7th term of an A.P. is equal to 11 times the 11th term, find the 18th term [5 marks]

Answer. Let the A.P. have first term a and common difference d . Then $T_r = a + (r - 1)d$.

$$7T_7 = 11T_{11} \Rightarrow 7(a + 6d) = 11(a + 10d).$$

So

$$7a + 42d = 11a + 110d \Rightarrow 4a + 68d = 0 \Rightarrow a = -17d.$$

The 18th term is

$$T_{18} = a + 17d = -17d + 17d = 0.$$

Final answer: $T_{18} = 0.$

1(e) If $1, \omega, \omega^2$ are cube roots of unity, find $(2 + \omega + \omega^2)^6 + (3 + \omega + \omega^2)^6$ [5 marks]

Answer. For cube roots of unity,

$$1 + \omega + \omega^2 = 0 \Rightarrow \omega + \omega^2 = -1.$$

Therefore

$$2 + \omega + \omega^2 = 1, \quad 3 + \omega + \omega^2 = 2.$$

Hence

$$(2 + \omega + \omega^2)^6 + (3 + \omega + \omega^2)^6 = 1^6 + 2^6 = 1 + 64 = 65.$$

Final answer: 65.

1(f) Find the quadratic equation with real coefficients and roots $\frac{m-n}{m+n}, -\frac{m+n}{m-n}$ [5 marks]

Answer. Let

$$\alpha = \frac{m-n}{m+n}, \quad \beta = -\frac{m+n}{m-n}.$$

Then

$$\alpha\beta = -1.$$

Also

$$\alpha + \beta = \frac{m-n}{m+n} - \frac{m+n}{m-n} = \frac{(m-n)^2 - (m+n)^2}{m^2 - n^2} = \frac{-4mn}{m^2 - n^2}.$$

The required equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0.$$

Thus

$$x^2 + \frac{4mn}{m^2 - n^2}x - 1 = 0.$$

Multiplying by $m^2 - n^2$,

$$(m^2 - n^2)x^2 + 4mnx - (m^2 - n^2) = 0.$$

Final answer: $(m^2 - n^2)x^2 + 4mnx - (m^2 - n^2) = 0.$

1(g) Evaluate $\int x\sqrt{3-2x} \, dx$

[5 marks]

Answer. Put $u = 3 - 2x$. Then $du = -2dx$, $dx = -du/2$, and $x = (3 - u)/2$.

$$\int x\sqrt{3-2x} \, dx = -\frac{1}{4} \int (3-u)u^{1/2} \, du.$$

So

$$= -\frac{1}{4} \left(3 \cdot \frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right) + C = -\frac{1}{2} u^{3/2} + \frac{1}{10} u^{5/2} + C.$$

Substituting $u = 3 - 2x$,

$$\int x\sqrt{3-2x} \, dx = \frac{1}{10}(3-2x)^{5/2} - \frac{1}{2}(3-2x)^{3/2} + C.$$

Equivalently,

$$= -\frac{x+1}{5}(3-2x)^{3/2} + C.$$

Final answer: $-\frac{x+1}{5}(3-2x)^{3/2} + C.$

1(h) A spherical balloon is inflated at $900 \text{ cm}^3/\text{s}$. How fast is radius increasing when $r = 25 \text{ cm}$?

[5 marks]

Answer. For a sphere,

$$V = \frac{4}{3}\pi r^3.$$

Differentiating with respect to t ,

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Given $dV/dt = 900$ and $r = 25$,

$$900 = 4\pi(25)^2 \frac{dr}{dt} = 2500\pi \frac{dr}{dt}.$$

Thus

$$\frac{dr}{dt} = \frac{900}{2500\pi} = \frac{9}{25\pi} \text{ cm/s}.$$

Final answer: $\frac{dr}{dt} = \frac{9}{25\pi} \text{ cm/s} \approx 0.115 \text{ cm/s}.$

Question 2

2(a) Solve by matrix method

[5 marks]

$$2x - y + 3z = 5, \quad 3x + 2y - z = 7, \quad 4x + 5y - 5z = 9.$$

Answer. The augmented matrix is

$$\left[\begin{array}{ccc|c} 2 & -1 & 3 & 5 \\ 3 & 2 & -1 & 7 \\ 4 & 5 & -5 & 9 \end{array} \right].$$

Solving the linear system by row reduction gives

$$x = \frac{43}{18}, \quad y = -\frac{1}{18}, \quad z = \frac{1}{18}.$$

Verification:

$$2\left(\frac{43}{18}\right) - \left(-\frac{1}{18}\right) + 3\left(\frac{1}{18}\right) = \frac{86 + 1 + 3}{18} = 5,$$

$$3\left(\frac{43}{18}\right) + 2\left(-\frac{1}{18}\right) - \frac{1}{18} = \frac{129 - 2 - 1}{18} = 7,$$

$$4\left(\frac{43}{18}\right) + 5\left(-\frac{1}{18}\right) - 5\left(\frac{1}{18}\right) = \frac{172 - 5 - 5}{18} = 9.$$

Final answer: $x = \frac{43}{18}, y = -\frac{1}{18}, z = \frac{1}{18}.$

2(b) Use mathematical induction to show $1 + 4 + 7 + \cdots + (3n - 2) = \frac{1}{2}n(3n - 1)$ [5 marks]

Answer. Let

$$P(n) : 1 + 4 + 7 + \cdots + (3n - 2) = \frac{1}{2}n(3n - 1).$$

For $n = 1$,

$$\text{LHS} = 1, \quad \text{RHS} = \frac{1}{2}(1)(3 - 1) = 1.$$

So $P(1)$ is true.

Assume $P(k)$ is true:

$$1 + 4 + \cdots + (3k - 2) = \frac{1}{2}k(3k - 1).$$

For $k + 1$,

$$\begin{aligned} 1 + 4 + \cdots + (3k - 2) + [3(k + 1) - 2] &= \frac{1}{2}k(3k - 1) + (3k + 1). \\ &= \frac{3k^2 - k + 6k + 2}{2} = \frac{3k^2 + 5k + 2}{2} = \frac{(k + 1)(3k + 2)}{2}. \end{aligned}$$

But

$$\frac{1}{2}(k + 1)(3(k + 1) - 1) = \frac{1}{2}(k + 1)(3k + 2).$$

Thus $P(k + 1)$ is true. Hence, by induction, the formula holds for all $n \in \mathbb{N}$.

Final answer: The identity is proved by mathematical induction.

2(c) Find quadratic equation with roots $\frac{1}{5 - \sqrt{72}}, \frac{1}{5 + 6\sqrt{2}}$

[5 marks]

Answer. Since $\sqrt{72} = 6\sqrt{2}$, the roots are

$$\alpha = \frac{1}{5 - 6\sqrt{2}}, \quad \beta = \frac{1}{5 + 6\sqrt{2}}.$$

Then

$$\alpha + \beta = \frac{(5 + 6\sqrt{2}) + (5 - 6\sqrt{2})}{25 - 72} = \frac{10}{-47} = -\frac{10}{47},$$

$$\alpha\beta = \frac{1}{(5 - 6\sqrt{2})(5 + 6\sqrt{2})} = \frac{1}{25 - 72} = -\frac{1}{47}.$$

The equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0.$$

Therefore

$$x^2 + \frac{10}{47}x - \frac{1}{47} = 0.$$

Multiplying by 47,

$$47x^2 + 10x - 1 = 0.$$

Final answer: $47x^2 + 10x - 1 = 0$.

2(d) If $\left(\frac{1-i}{1+i}\right)^{10} = a + ib$, show the values of a and b

[5 marks]

Answer. First simplify the base:

$$\frac{1-i}{1+i} = \frac{(1-i)^2}{(1+i)(1-i)} = \frac{1-2i+i^2}{2} = \frac{-2i}{2} = -i.$$

Thus

$$\left(\frac{1-i}{1+i}\right)^{10} = (-i)^{10} = (((-i)^4)^2)(-i)^2 = 1^2(-1) = -1.$$

Hence

$$a + ib = -1 + 0i.$$

Note: The printed question says to show $a = 1$ and $b = 0$, but direct calculation gives $a = -1$ and $b = 0$. Therefore the printed statement appears to contain a sign or exponent error.

Final answer: $a = -1, b = 0$ for the expression as printed.

Question 3

3(a) If $A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 5 & 2 \\ 2 & -1 & 7 \end{bmatrix}$, show that A is row equivalent to I_3 [5 marks]

Answer. A square matrix is row equivalent to I_3 if its determinant is non-zero. Here

$$\det A = \begin{vmatrix} 1 & 1 & 3 \\ 0 & 5 & 2 \\ 2 & -1 & 7 \end{vmatrix}.$$

Expanding along the first row,

$$\det A = 1(35 + 2) - 1(0 - 4) + 3(0 - 10) = 37 + 4 - 30 = 11.$$

Since $\det A = 11 \neq 0$, A is non-singular. Hence it can be reduced by elementary row operations to the identity matrix I_3 .

$$A \sim I_3.$$

Final answer: A is row equivalent to I_3 because $\det A = 11 \neq 0$.

3(b) Solve $\frac{2x-5}{x+2} < 5$, $x \in \mathbb{R}$, and graph the solution set [5 marks]

Answer. The expression is not defined at $x = -2$.

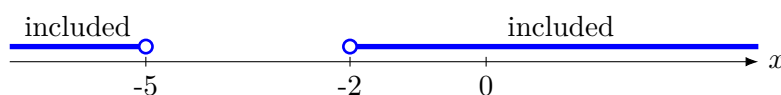
$$\begin{aligned} \frac{2x-5}{x+2} < 5 &\Rightarrow \frac{2x-5-5x-10}{x+2} < 0 \Rightarrow \frac{-3x-15}{x+2} < 0. \\ &\Rightarrow -3\frac{x+5}{x+2} < 0 \Rightarrow \frac{x+5}{x+2} > 0. \end{aligned}$$

Critical points are $x = -5$ and $x = -2$. Testing intervals gives

$$(-\infty, -5) : +, \quad (-5, -2) : -, \quad (-2, \infty) : +.$$

Therefore

$$x \in (-\infty, -5) \cup (-2, \infty).$$



Final answer: $(-\infty, -5) \cup (-2, \infty)$.

3(c) Solve $32x^3 - 48x^2 + 22x - 3 = 0$, given roots are in A.P. [5 marks]

Answer. Factorising the polynomial,

$$32x^3 - 48x^2 + 22x - 3 = (2x-1)(4x-1)(4x-3).$$

Therefore,

$$\begin{aligned} 2x-1 &= 0 \Rightarrow x = \frac{1}{2}, \\ 4x-1 &= 0 \Rightarrow x = \frac{1}{4}, \end{aligned}$$

$$4x - 3 = 0 \Rightarrow x = \frac{3}{4}.$$

The roots are

$$\frac{1}{4}, \frac{1}{2}, \frac{3}{4},$$

which are in A.P. with common difference $\frac{1}{4}$.

Final answer: $x = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$.

3(d) Determine the local maxima and local minima of $f(x) = x^3 - 6x^2 + 9x + 100$ [5 marks]

Answer.

$$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3).$$

Critical points are $x = 1$ and $x = 3$.

$$f''(x) = 6x - 12.$$

At $x = 1$,

$$f''(1) = 6 - 12 = -6 < 0,$$

so $x = 1$ gives local maximum.

$$f(1) = 1 - 6 + 9 + 100 = 104.$$

At $x = 3$,

$$f''(3) = 18 - 12 = 6 > 0,$$

so $x = 3$ gives local minimum.

$$f(3) = 27 - 54 + 27 + 100 = 100.$$

Final answer: Local maximum: $(1, 104)$. Local minimum: $(3, 100)$.

Question 4

4(a) Check continuity at $x = 0$ for $f(x) = \begin{cases} \frac{2|x|}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ [5 marks]

Answer. For $x > 0$, $|x| = x$, so

$$f(x) = \frac{2x}{x} = 2.$$

Thus

$$\lim_{x \rightarrow 0^+} f(x) = 2.$$

For $x < 0$, $|x| = -x$, so

$$f(x) = \frac{2(-x)}{x} = -2.$$

Thus

$$\lim_{x \rightarrow 0^-} f(x) = -2.$$

Since the left-hand and right-hand limits are not equal, $\lim_{x \rightarrow 0} f(x)$ does not exist. Also $f(0) = 0$.

Final answer: $f(x)$ is not continuous at $x = 0$.

4(b) Find vector and cartesian equations of the line through $(1, -1, -2)$ and parallel to $3\hat{i} - 2\hat{j} + 5\hat{k}$ [5 marks]

Answer. The point is $(1, -1, -2)$ and the direction vector is $(3, -2, 5)$.

The vector equation is

$$\vec{r} = (\hat{i} - \hat{j} - 2\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 5\hat{k}), \quad \lambda \in \mathbb{R}.$$

The parametric equations are

$$x = 1 + 3\lambda, \quad y = -1 - 2\lambda, \quad z = -2 + 5\lambda.$$

Hence the cartesian equation is

$$\frac{x-1}{3} = \frac{y+1}{-2} = \frac{z+2}{5}.$$

Final answer: $\vec{r} = (\hat{i} - \hat{j} - 2\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 5\hat{k})$ and $\frac{x-1}{3} = \frac{y+1}{-2} = \frac{z+2}{5}$.

4(c) Determine the length of the curve $y = 2x^{3/2}$ from $(1, 2)$ to $(4, 16)$ [5 marks]

Answer. The arc length is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

Given

$$y = 2x^{3/2} \Rightarrow \frac{dy}{dx} = 3x^{1/2} = 3\sqrt{x}.$$

Therefore

$$L = \int_1^4 \sqrt{1 + 9x} dx.$$

Let $u = 1 + 9x$, so $du = 9dx$.

$$L = \frac{1}{9} \int_{10}^{37} u^{1/2} du = \frac{1}{9} \left[\frac{2}{3} u^{3/2} \right]_{10}^{37} = \frac{2}{27} (37^{3/2} - 10^{3/2}).$$

Final answer: $L = \frac{2}{27} (37^{3/2} - 10^{3/2})$.

4(d) Find the sum of all integers between 100 and 1000 divisible by 9 [5 marks]

Answer. The first multiple of 9 greater than 100 is 108 and the last multiple of 9 less than 1000 is 999. This is an A.P.:

$$108, 117, 126, \dots, 999$$

with $a = 108$, $d = 9$, and $l = 999$.

$$l = a + (n - 1)d \Rightarrow 999 = 108 + (n - 1)9.$$

$$891 = 9(n - 1) \Rightarrow n - 1 = 99 \Rightarrow n = 100.$$

Sum is

$$S_n = \frac{n}{2}(a + l) = \frac{100}{2}(108 + 999) = 50 \times 1107 = 55350.$$

Final answer: 55350.

Question 5

5(a) Maximize $5x + 2y$ subject to the given constraints

[5 marks]

$$-2x - 3y \leq -6, \quad x - 2y \leq 2, \quad 6x + 4y \leq 24, \quad -3x + 2y \leq 3, \quad x \geq 0, \quad y \geq 0.$$

Answer. Convert the first and third constraints:

$$2x + 3y \geq 6, \quad 3x + 2y \leq 12.$$

The boundary lines are

$$2x + 3y = 6, \quad x - 2y = 2, \quad 3x + 2y = 12, \quad -3x + 2y = 3.$$

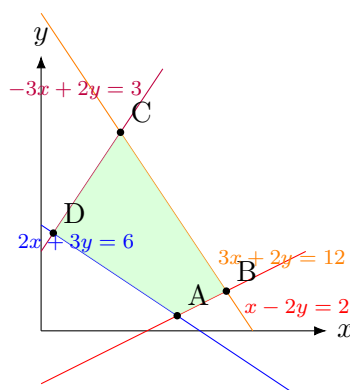
The feasible corner points are obtained by intersecting boundary lines:

$$\left(\frac{18}{7}, \frac{2}{7}\right), \quad \left(\frac{3}{13}, \frac{24}{13}\right), \quad \left(\frac{7}{2}, \frac{3}{4}\right), \quad \left(\frac{3}{2}, \frac{15}{4}\right).$$

Evaluate $Z = 5x + 2y$:

Corner point	$Z = 5x + 2y$
$\left(\frac{18}{7}, \frac{2}{7}\right)$	$\frac{94}{7}$
$\left(\frac{3}{13}, \frac{24}{13}\right)$	$\frac{63}{13}$
$\left(\frac{7}{2}, \frac{3}{4}\right)$	19
$\left(\frac{3}{2}, \frac{15}{4}\right)$	15

The maximum is 19 at $\left(\frac{7}{2}, \frac{3}{4}\right)$.



Final answer: Maximum value is 19 at $\left(\frac{7}{2}, \frac{3}{4}\right)$.

5(b) Find the area bounded by $y = x^2$ and $y^2 = x$

[5 marks]

Answer. The curve $y^2 = x$ gives $y = \sqrt{x}$ for the upper branch in the first quadrant. Intersections satisfy

$$y = x^2, \quad x = y^2.$$

Substitute $x = y^2$ into $y = x^2$:

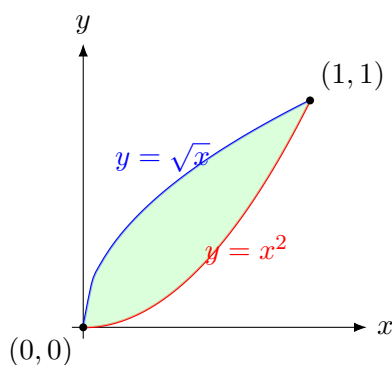
$$y = (y^2)^2 = y^4 \Rightarrow y(y^3 - 1) = 0.$$

So $y = 0, 1$, giving intersection points $(0, 0)$ and $(1, 1)$.

For $0 \leq x \leq 1$, upper curve is $y = \sqrt{x}$ and lower curve is $y = x^2$. Therefore area is

$$A = \int_0^1 (\sqrt{x} - x^2) dx.$$

$$A = \left[\frac{2}{3} x^{3/2} - \frac{1}{3} x^3 \right]_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}.$$



Final answer: Area = $\frac{1}{3}$ square unit.

5(c) Reduce $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ to normal form and determine rank [5 marks]

Answer. The determinant is

$$\det A = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix}.$$

Expanding along the first row,

$$\det A = -1 \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + 2 \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -(1 - 9) + 2(1 - 6) = 8 - 10 = -2.$$

Since $\det A = -2 \neq 0$, the matrix is non-singular and row-column equivalent to I_3 . Thus its normal form is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Hence the rank is 3.

Final answer: Normal form = I_3 and $\text{rank}(A) = 3$.

5(d) If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, show that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$ [5 marks]

Answer. Given

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}.$$

So

$$\vec{c} = -(\vec{a} + \vec{b}).$$

Now

$$\vec{b} \times \vec{c} = \vec{b} \times [-(\vec{a} + \vec{b})] = -\vec{b} \times \vec{a} - \vec{b} \times \vec{b}.$$

Since $\vec{b} \times \vec{b} = \vec{0}$ and $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$,

$$\vec{b} \times \vec{c} = \vec{a} \times \vec{b}.$$

Also

$$\vec{c} \times \vec{a} = [-(\vec{a} + \vec{b})] \times \vec{a} = -\vec{a} \times \vec{a} - \vec{b} \times \vec{a} = \vec{a} \times \vec{b}.$$

Therefore

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}.$$

Final answer: The required vector product identity is proved.

Last Page: Theory and Formula Sheet

Determinants and matrices. For a square matrix A , if $\det A \neq 0$, then A is non-singular and row equivalent to I . A quadratic equation with roots α, β is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$. For matrix powers, first simplify the matrix expression and then multiply.

A.P. and series. The n th term of an A.P. is $T_n = a + (n-1)d$. The sum of an A.P. is $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$. For a finite G.P., $a + ar + \dots + ar^{n-1} = a \frac{1-r^n}{1-r}$.

Complex numbers. For cube roots of unity, $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$. Also $i^2 = -1$, $i^4 = 1$. Rationalize complex fractions by multiplying numerator and denominator by the conjugate.

Calculus. For integration by substitution, put the inner expression equal to u . Related rates use differentiation with respect to time. For a sphere, $V = \frac{4}{3}\pi r^3$ and $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. Arc length is $L = \int_a^b \sqrt{1 + (dy/dx)^2} dx$.

Continuity and extrema. A function is continuous at a if $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$. For local maxima/minima, solve $f'(x) = 0$. If $f''(a) < 0$, local

maximum; if $f''(a) > 0$, local minimum.

Induction. To prove $P(n)$: prove $P(1)$, assume $P(k)$, then prove $P(k+1)$. This proves the result for all $n \in \mathbb{N}$.

Inequalities. Move all terms to one side, factor, identify critical points, and use a sign chart. Denominator zeros are excluded from the solution set.

Coordinate geometry and vectors. Line through point $\vec{r}_0$ parallel to vector $\vec{v}$ is $\vec{r} = \vec{r}_0 + \lambda \vec{v}$. Cartesian form: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$. If $\vec{a} + \vec{b} + \vec{c} = 0$, then substitute one vector as negative sum of the other two.

Linear programming. Convert each inequality into a boundary line. Find the feasible region, calculate all corner points, and evaluate the objective function at each corner. Maximum/minimum occurs at a corner point for a bounded feasible polygon.

Area between curves. Area is $\int (\text{upper curve} - \text{lower curve}) dx$ or $\int (\text{right curve} - \text{left curve}) dy$ after finding intersection limits.